

as shown in Fig. 4. If the LED is forward-biased, its light falls directly on the base of the photo transistor and it conducts. But, if any obstruction like a thin strip, card paper or post card is placed in the gap, the light does not fall and the photo transistor does not conduct.

DC motor driver IC L293D.

L293D (IC2) contains two in-built H-bridge driver circuits. In its common mode of operation, two DC motors can be driven simultaneously, both in forward and reverse directions. The operation of the two motors can be controlled by input logic at pins 2 and 7, and 10 and 15. Input logic 00 or 11 will stop the corresponding motor. Logic 01 and 10 will rotate it in clockwise and anti-clockwise directions, respectively. In the circuit, it is programmed to run 5rpm DC motor in a clockwise as well as anti-clockwise direction. In this circuit, the input logics are fed to pins 2 and 7 to control the motor connected across pins 3 and 6 of IC2.

LCD panel. The 20x4 LCD displays the scanning angle, measured distance, messages like Object Detected and Scanning, etc. It accepts ASCII value of any letter or digit and displays it on the selected line.

LED dial. Twelve red-colour LEDs are arranged like a dial of clock. Each LED indicates an angle between 0° and 330° that is separated from the next or previous by 30 degrees. When an object is detected, its angle is indicated by the corresponding angle of LED. This gives an approximate idea of position, direction and angle of an object with respect to reference position.

Buzzer. It is a piezoelectric DC buzzer that generates audible beep-like sounds when given DC supply of 5V. It is used to give short beep sounds on detection of an object.

Opto interrupt sensor circuit.

This circuit generates a negative



Fig. 3: Ultrasonic distance sensor—serial out module

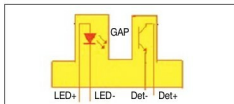


Fig. 4: Slotted opto isolator module MOC7811

pulse when the strip passes the gap in the sensor. This pulse is used to generate an interrupt signal for the microcontroller. As shown in Fig. 2, the circuit comprises a slotted opto isolator module, BC547 npn transistor T1 and a few resistors.

The internal LED is forward-biased through the current-limiting resistor of 330Ω. This turns it on continuously. The falling LED light drives the photo transistor to saturation. Its output at the collector is low, which drives T1 into cut-off. This means that output of the circuit is high, which is the normal state when the falling light is not interrupted. When a strip passes through the gap, it blocks the light and the photo transistor goes into cut-off. Its output at the collector is high, which drives transistor T1 into saturation and gives low output at its collector.

As the strip passes through the gap, we get high to low, which is taken as interrupt. Interconnection of various components with microcontroller ATmega128 (IC1) is shown in Fig. 2 and described below.

1. Serial data output pin Tx of the sensor is connected to PEO of IC3.

2. Opto interrupt sensor interfacing circuit is connected to external interrupt 4 pin at pin 6 of IC3.

3. Pins PB5 and PB6 of PORTB are connected to inputs (IN1 and IN2) of L293D chip. These pins are used to rotate the motor clockwise and anti-clockwise. Pin PB4 is connected to enable input of L293D chip, which is used to enable the output of the chip.

4. DC motor is connected to two outputs (OUT1 and OUT2) of L293D (IC2).

5. Pin PF0 of PORTF is connected to operation (scanning) LED17 through current limiting resistor R19. Another pin, PF1, is connected to the buzzer through npn transistor (T2) BC547.

6. PORTA pins PA4 through PA7 are connected to data pins D4 through D7 of the LCD. Pins PA0 and PA2 are connected to control pins RS and EN of the LCD, respectively. R/W pin of LCD is connected to ground.

7. LCD pins 2 and 15 are connected to Vcc, the latter through 100-ohm resistor R20, and pins 1 and 16 are connected to ground. Pin 3 of the LCD, which is used for contrast control, is connected with 10k pot, as shown in the figure.

8. PORTC pins PC0 through PC7 are connected to cathodes of eight LEDs. Anodes of all the LEDs are connected to Vcc through current-limiting resistors.

9. Similarly, four PORTG pins PG0 through PG3 are connected to four LEDs.

10. The 12MHz crystal with two 22pF capacitors is connected to pins XTAL1 and XTAL2.

11. The complete system works on 5V DC supply, with additional 12V DC supply given to pin 8 of L293D (IC2) for the DC motor.

System operation

1. Initially, the motor is stationary and the strip is in the gap of the slotted opto isolator module. This is the reference angle 0° (position) for ultrasonic sensor module.

2. Next, the motor starts rotating in a clockwise or anti-clockwise direction. The speed of the motor is